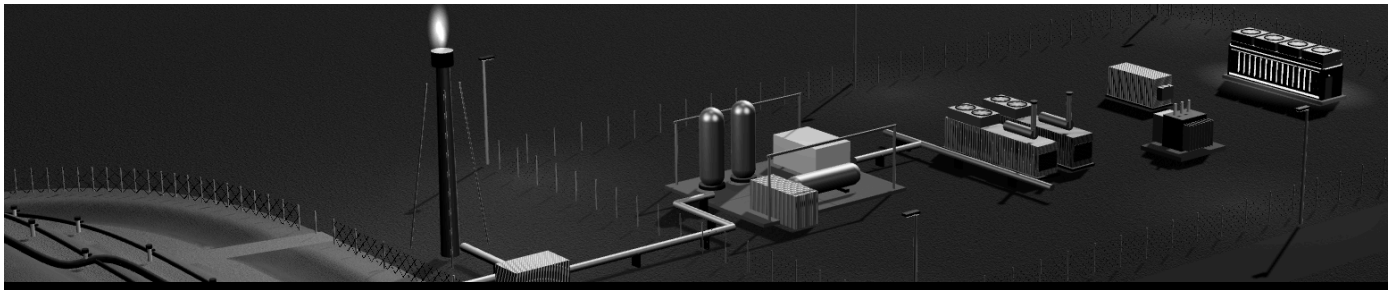
City of Richmond Wastewater Treatment Plant, Richmond, Virginia: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

1.0 MW	1	6 ac	91k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $273k	$6k to $12k	$6k to $285k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=300&CH4=60

THE MATH

Digester gas, low estimate	300 scfm
Methane (assumed)	60%
Energy in the gas	10.9 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 1.0 MW
Compute (power less cooling and losses, /1.2)	about 0.8 MW
Compute modules (about 1 MW each)	1
Gas engines (running plus one spare)	2 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 6 acres
Gas bought per year (95% run time)	91k MMBtu
Gas payment, up to $3.00/MMBtu	up to $273k
Land lease at $1,000 to $2,000 per acre per year	$6k to $12k

Biogas: methane at about 1,012 Btu per cubic foot. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Flow is an estimate, not a measurement
- How much gas is flared or used for heat today
- Space for a parcel of one to two acres

Source: No gas figure published. Estimate: about 6.9 scfm of digester gas per MGD (EPA CHP Partnership rule of thumb) at about 61.6 MGD gives about 425 scfm; this first read takes about 300 scfm, leaving the rest for digester heat. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.